Power Series

Ratio Test:

If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|<1$, then $\sum_{n=1}^{\infty}a_n$ converges.

If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|>1$, then $\sum_{n=1}^{\infty}a_n$ diverges.

If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=1$, then use another test.

1) If $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right|=0$, then powers series converges for all x.

Interval of convergence = $(-\infty,\infty)$. Radius of convergence = ∞.

2) If $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right|=\infty$, then powers series only converges for $x=0$.

Interval of convergence = $\{0\}$. Radius of convergence = 0

1) Power Series: $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n+4}$

Let $u_n = (-1)^n \frac{x^n}{n+4}$ and $u_{n+1} = (-1)^{n+1} \frac{x^{n+1}}{(n+1)+4} = (-1)^{n+1} \frac{x^{n+1}}{n+5}$

a) $\lim_{n\to\infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^{n+1} \frac{x^{n+1}}{n+5}}{(-1)^n \frac{x^n}{n+4}} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{x^n} \right| \cdot \lim_{n\to\infty} \left| \frac{n+4}{n+5} \right| =$ __?__ (Multiple Choice)

A) $|x|$ B) $2|x|$ C) $3|x|$ D) $4|x|$

b) Find Interval of convergence = __?____________

2) Power Series: $\sum_{n=1}^{\infty} \frac{(4x)^n}{n^2}$

Let $u_n = \frac{(4x)^n}{n^2}$ and $u_{n+1} = \frac{(4x)^{n+1}}{(n+1)^2}$

a) $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{(4x)^{n+1}}{(n+1)^2}}{\frac{(4x)^n}{n^2}}\right| = \lim_{n\to\infty}\left|\frac{(4x)^{n+1}}{(4x)^n}\right| \cdot \lim_{n\to\infty}\left|\frac{n^2}{(n+1)^2}\right| = \underline{\quad ? \quad}$ (Multiple Choice)

A) $2|4x|$ B) $|4x|$ C) $3|5x|$ D) $3|x|$

b) Find Interval of convergence = $\underline{\quad ? \qquad\qquad\qquad}$

3) Power Series: $\sum_{n=1}^{\infty} \frac{x^{2n}}{(2n)!}$

Let $u_n = \frac{x^{2n}}{(2n)!}$ and $u_{n+1} = \frac{x^{2(n+1)}}{(2(n+1))!} = \frac{x^{2n+2}}{(2n+2)!}$

Hint: $x^{2n+2} = x^{2n}x^2$; $(2n+2)! = (2n+2)(2n+1)(2n)(2n-1)\cdots(3)(2)(1)$

$(2n)! = (2n)(2n-1)(2n-2)(2n-3)\cdots(3)(2)(1)$

a) $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{x^{2n+2}}{(2n+2)!}}{\frac{x^{2n}}{(2n)!}}\right| = \lim_{n\to\infty}\left|\frac{x^{2n+2}}{x^{2n}}\right| \lim_{n\to\infty}\left|\frac{(2n)!}{(2n+2)!}\right| = \underline{\quad ? \quad}$ (Multiple Choice)

A) $|x|$ B) 0 C) $|x|\cdot(n+2)$ D) ∞

b) Find Interval of convergence = $\underline{\quad ? \qquad\qquad}$

4) Power Series: $\sum_{n=1}^{\infty}(2n)!\left(\frac{x}{3}\right)^n$

Let $u_n=(2n)!\left(\frac{x}{3}\right)^n$ and $u_{n+1}=(2(n+1))!\left(\frac{x}{3}\right)^{n+1}=(2n+2)!\left(\frac{x}{3}\right)^{n+1}$

a) $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right|=\lim_{n\to\infty}\left|\frac{(2n+2)!\left(\frac{x}{3}\right)^{n+1}}{(2n)!\left(\frac{x}{3}\right)^n}\right|=\lim_{n\to\infty}\left|\frac{\left(\frac{x}{3}\right)^{n+1}}{\left(\frac{x}{3}\right)^n}\right|\cdot\lim_{n\to\infty}\left|\frac{(2n+2)!}{(2n)!}\right|=$ __?__ (Multiple Choice)

Hint: $\left(\frac{x}{3}\right)^{n+1}=\left(\frac{x}{3}\right)^n\left(\frac{x}{3}\right)^1$; $(2n+2)!=(2n+2)(2n+1)(2n)(2n-1)\cdots(3)(2)(1)$

$(2n)!=(2n)(2n-1)(2n-2)(2n-3)\cdots(3)(2)(1)$

A) $\left|\frac{x}{3}\right|$ B) 0 C) $\left|\frac{x}{3}\right|\cdot(n+2)(n+1)$ D) ∞

b) Find Interval of convergence = __?________________

5) Power Series: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{6^n}$

Let $u_n = \frac{(-1)^{n+1} x^n}{6^n}$ and $u_{n+1} = \frac{(-1)^{n+2} x^{n+1}}{6^{n+1}}$

a) $\lim_{n\to\infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n\to\infty} \left| \frac{\frac{(-1)^{n+2} x^{n+1}}{6^{n+1}}}{\frac{(-1)^{n+1} x^n}{6^n}} \right| = \lim_{n\to\infty} \left| \frac{x^{n+1}}{x^n} \right| \cdot \lim_{n\to\infty} \left| \frac{6^n}{6^{n+1}} \right| =$ ___?___ (Multiple Choice)

A) $|x|$ B) $|x| \cdot \frac{1}{6}$ C) $6|x|$ D) ∞

b) Find Interval of convergence = ___?___

6) Power Series: $\sum_{n=1}^{\infty} \frac{(x-3)^{n-1}}{3^{n-1}}$

Let $u_n = \frac{(x-3)^{n-1}}{3^{n-1}}$ and $u_{n+1} = \frac{(x-3)^n}{3^n}$

a) $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{(x-3)^n}{3^n}}{\frac{(x-3)^{n-1}}{3^{n-1}}}\right| = \lim_{n\to\infty}\left|\frac{(x-3)^n}{(x-3)^{n-1}}\right| \cdot \lim_{n\to\infty}\left|\frac{3^{n-1}}{3^n}\right| =$ ___?____ (Multiple Choice)

A) $|x-3|$ B) $|x-3|\cdot\frac{1}{3}$ C) $3|x-3|$ D) ∞

b) Find Interval of convergence = __?____________

7) Power Series: $\sum_{n=1}^{\infty} \frac{n}{n+1}(-2x)^{n-1}$

Let $u_n = \frac{n}{n+1}(-2x)^{n-1}$ and $u_{n+1} = \frac{n+1}{n+2}(-2x)^n$

a) $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{n+1}{n+2}(-2x)^n}{\frac{n}{n+1}(-2x)^{n-1}}\right| = \lim_{n\to\infty}\left|\frac{\frac{n+1}{n+2}}{\frac{n}{n+1}}\right| \cdot \lim_{n\to\infty}\left|\frac{(2x)^n}{(2x)^{n-1}}\right| =$ __?___(Multiple Choice)

A) $|2x|$ B) $2|2x|$ C) $3|3x|$ D) 0

b) Find the interval of convergence = __?______________

8) Power Series: $\sum_{n=1}^{\infty} \frac{x^{3n+1}}{(3n+1)!}$

Let $u_n = \frac{x^{3n+1}}{(3n+1)!}$ and $u_{n+1} = \frac{x^{3(n+1)+1}}{(3(n+1)+1)!} = \frac{x^{3n+4}}{(3n+4)!}$

a) $\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{x^{3n+4}}{(3n+4)!}}{\frac{x^{3n+1}}{(3n+1)!}}\right| = \lim_{n\to\infty}\left|\frac{x^{3n+4}}{x^{3n+1}}\right| \cdot \lim_{n\to\infty}\left|\frac{(3n+1)!}{(3n+4)!}\right| = \underline{\quad ? \quad}$ (Multiple Choice)

Hint: $x^{3n+4} = x^{3n}x^4$; $(3n+1)! = (3n+1)(3n)(3n-1)(3n-2)(3n-3)\cdots(3)(2)(1)$

$(3n+4)! = (3n+4)(3n+3)(3n+2)(3n+1)(3n)\cdots(3)(2)(1)$

A) $|x^3|\cdot(3n+2)$ B) $|x^3|\cdot(n)$ C) 0 D) ∞

b) Find the interval of convergence = $\underline{\quad ? \qquad\qquad\qquad}$